Tutorial 2
Relational algebra and basic SQL

Week 6
Tutor: Chen Junan

Question 1. A university course database

Throughout this question use the following schema (primary keys are underlined). In `Enrolled`, `grade` is NULL until the course has been assessed; in `Teaches` a tuple means that a course is offered in a term and taught by an instructor; a term is a string such as `'2025F'`.

```
Student(sid, sname, major, level)
Course(cid, title, dept, credits)
Enrolled(sid, cid, term, grade)
Instructor(iid, iname, dept)
Teaches(iid, cid, term)
```

(a) Write a relational algebra expression for each of the following.

(i) Titles of all courses in the `'CS'` department that are worth more than 3 credits.

(ii) Names of the students who have taken at least one course taught by instructor `'I1'`.

(iii) Names of the students who have taken a course in the `'CS'` department *or* in the `'Math'` department, and then the names of those who have taken a course in *both* (use a set union, then a set intersection).

(iv) Names of the students who have never taken any course.

(v) Names of the students who have taken *every* course in the `'CS'` department.

(vi) For each department, the number of courses and the average number of credits.

(b) Write an SQL query for each of the following.

(i) The names and levels of all students whose major is `'CS'` or `'Math'`.

(ii) For each enrolment, the student's name, the course title and the term.

(iii) The distinct titles of the courses taught by instructor `'I1'`.

(iv) The student IDs of the students who have at least one enrolment without a grade; and explain in one sentence why the condition `grade <> 'A'` does not return those rows.